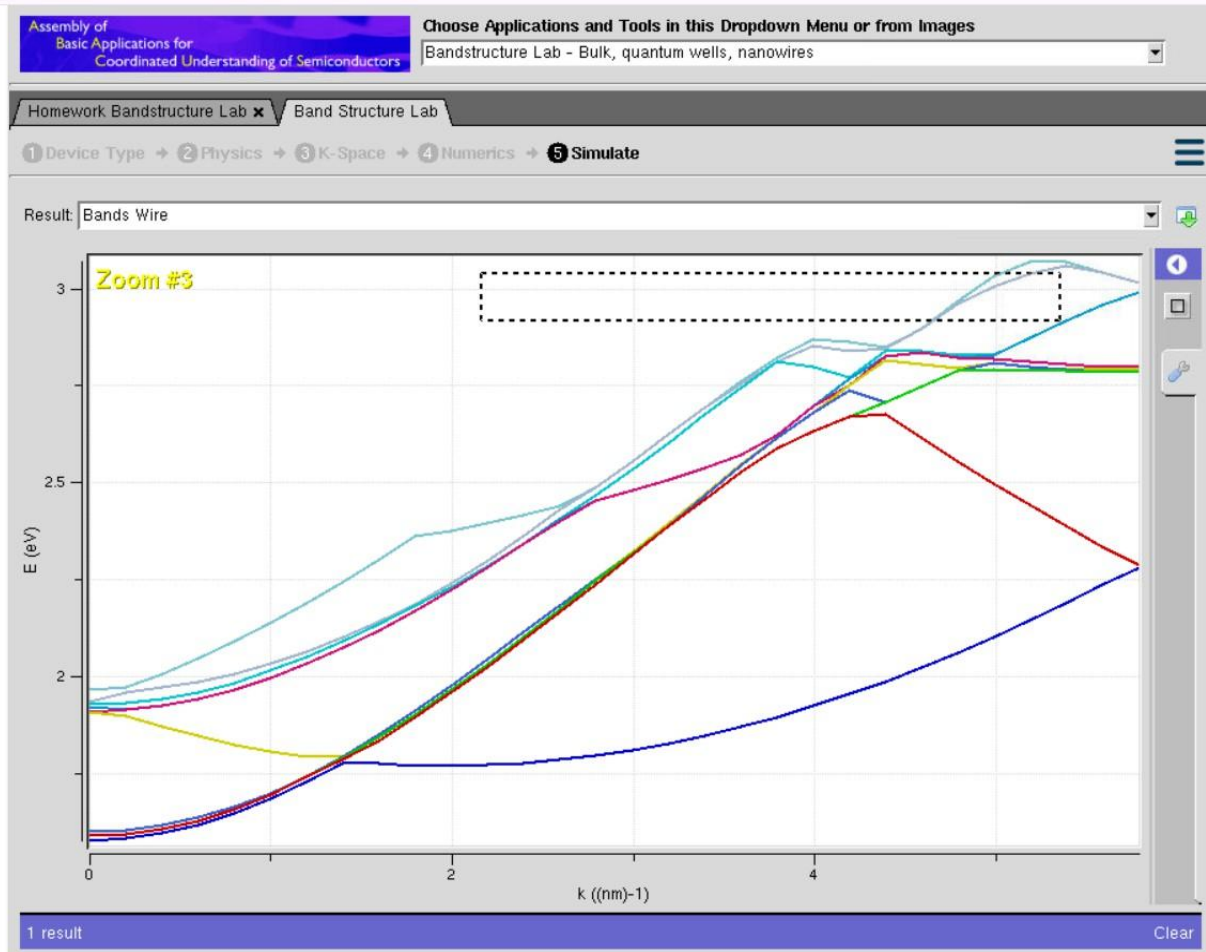


Exp. 4. Band Structure Simulation for Graphene and CNTs using NanoHub

V. Parkavi
192412241

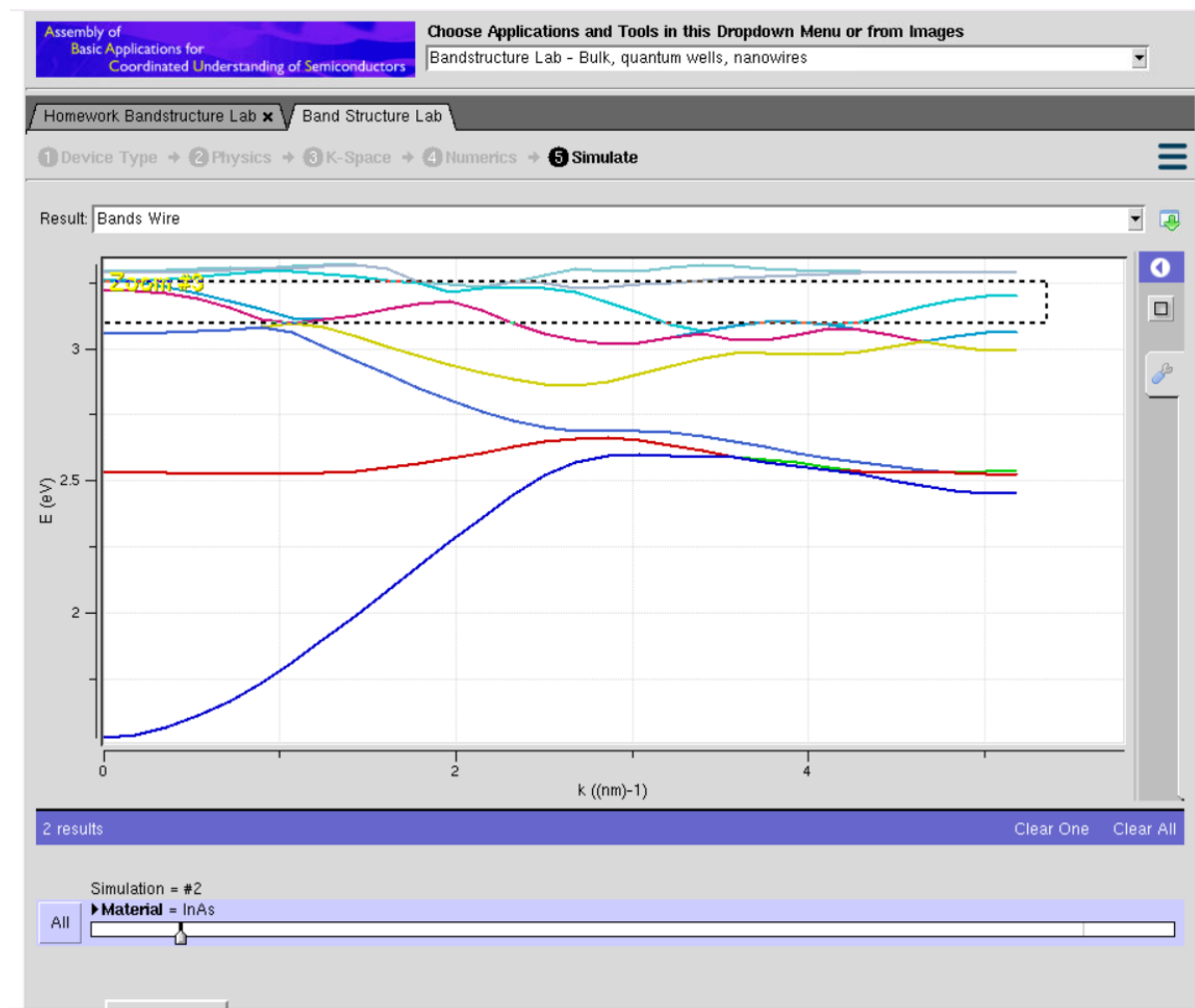
Aim: To simulate density of states of quantum well structure with various nano meter thickness.



This graph illustrates the relationship between **Energy (E, eV)** and **Wavevector (k, nm⁻¹)** for a **1D semiconductor nanowire** simulated using the **ABACUS tool in nanoHUB**. Due to quantum confinement in two spatial directions, the continuous energy bands of the bulk material split into discrete one-dimensional subbands.

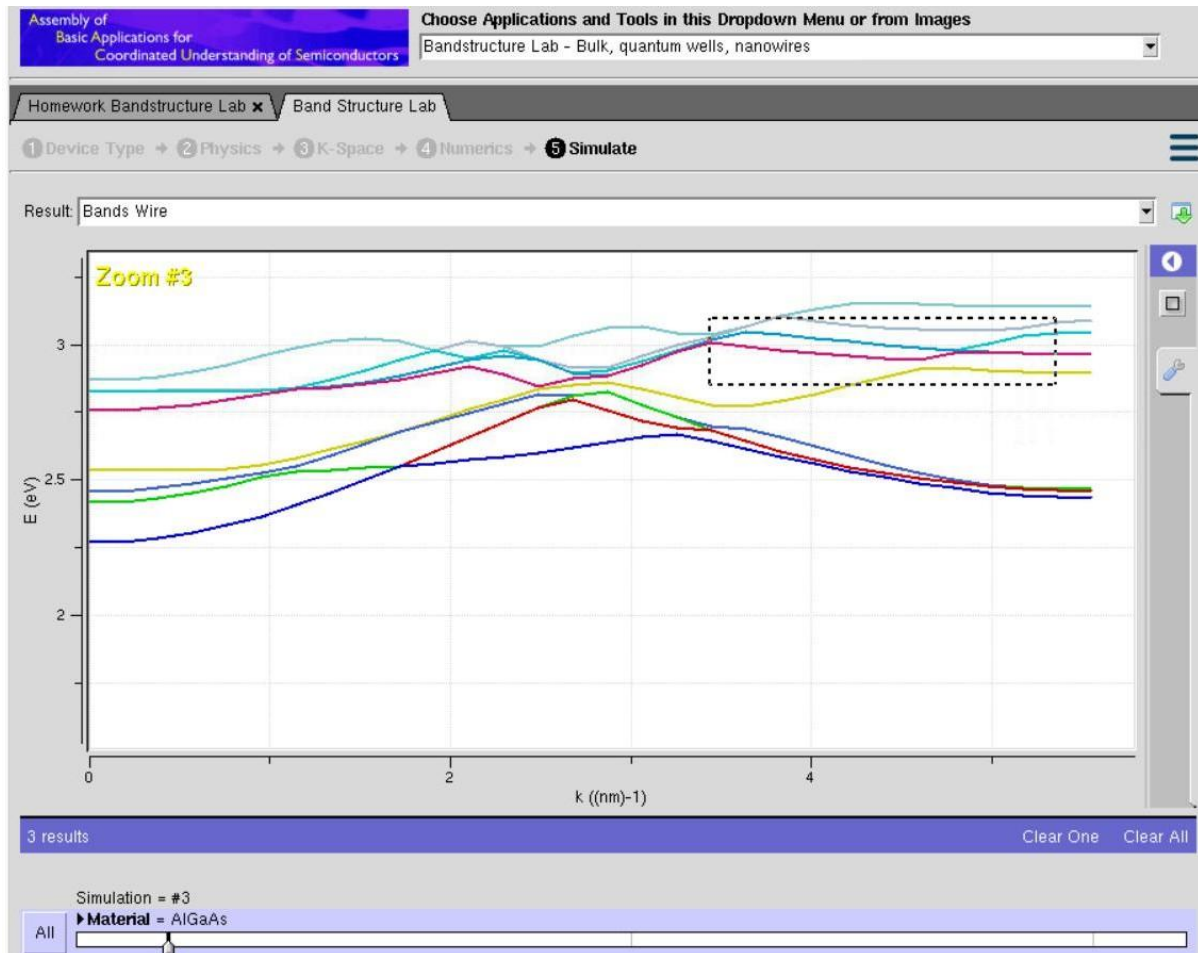
The curvature of the energy bands indicates the **effective mass** of charge carriers. Steeper parabolic bands correspond to a lower effective mass, allowing electrons to move more rapidly, whereas flatter bands, particularly near the upper-right region above **2.8 eV**, indicate a higher effective mass and an increased density of states. The subband minima at **k = 0** (approximately **1.8 eV**) define the onset of conduction, while the band intersections observed around **k = 3–4 nm⁻¹** suggest possible inter-subband scattering, which plays a significant role in understanding electron transport in nanoscale semiconductor devices.

Case Study 1: Simulate the band-structure of InAs circular nanowire



Case Study 1 :examines the band structure of an **Indium Arsenide (InAs) circular nanowire** using the **ABACUS Band Structure Lab** available on **nanoHUB**. The objective is to investigate the influence of the material's low electron effective mass under one-dimensional quantum confinement. This study is important for the development of high-electron-mobility nanoscale transistors, where quantum confinement strongly affects carrier transport. The simulation is performed by defining a **1D circular nanowire geometry**, selecting **InAs** as the semiconductor material, and computing the **energy–wavevector (E–k) dispersion relation**. The resulting band structure exhibits **widely separated, strongly curved energy subbands**, reflecting the light effective mass and enhanced electron mobility characteristic of InAs.

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



Case Study 2: investigates the band structure of an **Aluminum Gallium Arsenide (AlGaAs) circular nanowire** using the **ABACUS tool** on **nanoHUB**. The study focuses on how variations in the **molar fraction** and the material's relatively higher electron effective mass influence the **bandgap** and the distribution of energy subbands. This analysis is important for the design of **heterostructure nanowires** and **quantum-based optoelectronic devices**. The simulation is carried out by selecting a **1D circular nanowire geometry**, choosing **AlGaAs** as the semiconductor material, assigning the required **molar fraction**, and calculating the **electronic band structure**. The results show **closely spaced and densely packed energy subbands**, which are characteristic of the higher carrier effective mass in AlGaAs.

In conclusion, reducing the dimensions of a semiconductor to form a **one-dimensional (1D) nanowire** results in strong **quantum confinement**, which transforms continuous bulk energy bands into discrete quantized subbands. By simulating the **Energy–Wavevector (E–k) band structure** using **nanoHUB**, researchers can better understand and tailor these subband characteristics. This insight is essential for improving carrier transport and enhancing the performance of future **nanoscale electronic and optoelectronic devices**.

